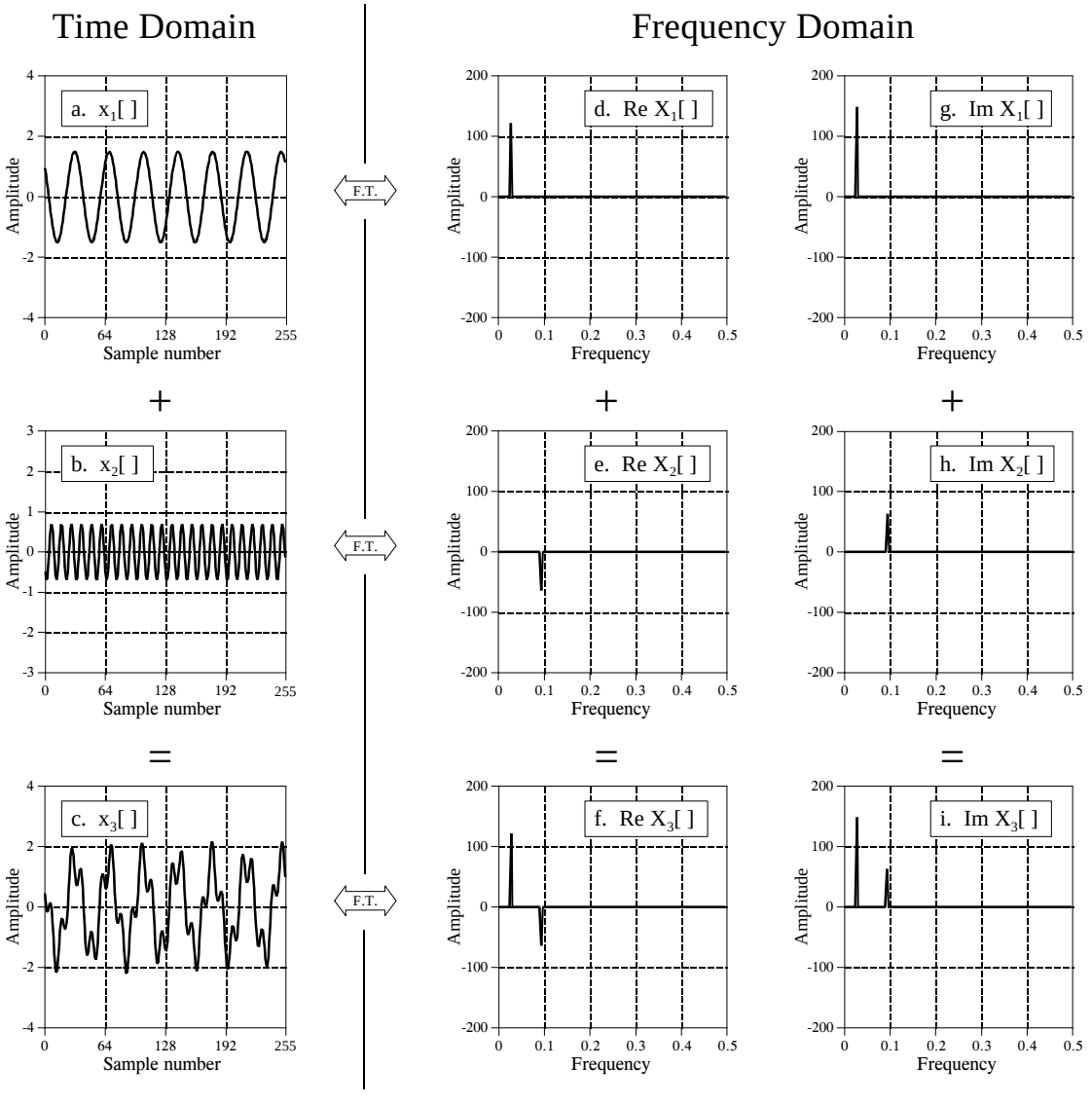

FIGURE 10-2
Additivity of the Fourier transform. Adding two or more signals in one domain results in the corresponding signals being added in the other domain. In this illustration, the time domain signals in (a) and (b) are added to produce the signal in (c). This results in the corresponding real and imaginary parts of the frequency spectra being added.

polar form. This can also be understood in terms of how sinusoids behave. Imagine adding two sinusoids having the same frequency, but with different amplitudes (A_1 and A_2) and phases (ϕ_1 and ϕ_2). If the two phases happen to be same ($\phi_1 = \phi_2$), the amplitudes will add ($A_1 + A_2$) when the sinusoids are added. However, if the two phases happen to be exactly opposite ($\phi_1 = -\phi_2$), the amplitudes will *subtract* ($A_1 - A_2$) when the sinusoids are added. The point is, when sinusoids (or spectra) are in polar form, they *cannot* be added by simply adding the magnitudes and phases.